

Statistical Arbitrage Early Warning System: LSTM-based Anomaly Detection on Bloomberg Macro Data

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New Technologies for Finance — GSoM Hackathon • *Bloomberg Weekly Dataset, Jan 2000 – Apr 2021*

Abstract

We develop a pairs-trading Early Warning System (EWS) that detects anomalous spread behaviour in weekly Bloomberg macro data. Candidate pairs are identified through OPTICS clustering on PCA-reduced returns, followed by a sequential cointegration filter. An LSTM novelty detector, trained exclusively on risk-on weeks, learns normal spread dynamics and flags deviations as potential stress events. Compared against a classical $\pm 2\sigma$ threshold baseline on a COVID-19 stress test (Jan 2020 – Apr 2021), the LSTM achieves a portfolio Sharpe ratio of 2.596 and an average ROI of +10.23%, outperforming the threshold model by +1.666 Sharpe points.

Problem Statement & Dataset

The goal is to detect regime breaks in financial spread data, framed as novelty detection: identify weeks where the spread between two instruments diverges anomalously from its historical norm.

Dataset. The dataset contains 1,111 weekly observations (Jan 2000 – Apr 2021) across 42 Bloomberg macro features including bond indices, equities, FX, rates, and commodities, together with a binary label $Y \in \{0, 1\}$ indicating risk-off episodes. The dataset is imbalanced: $Y = 0$ (risk-on) accounts for 874 weeks (79%) and $Y = 1$ (risk-off) for 237 weeks (21%). There are no missing values. Primary metric is F1 on the anomaly class $Y = 1$; accuracy is misleading at $\sim 21\%$ prevalence.

Pair Selection

With 42 features there are ~ 861 possible pairs. To restrict the search to economically coherent instruments, we first apply **OPTICS** (Ordering Points To Identify the Clustering Structure) on PCA-reduced weekly returns (10 components, 99.7% variance explained). OPTICS identifies 2 clusters of 5 features each and excludes 32 noise features as outliers, reducing the search space from 861 to 20 intra-cluster candidate pairs.

Each candidate is then filtered through four sequential criteria: (1) non-stationarity of both legs (ADF $p > 0.10$); (2) cointegration of the spread (Engle–Granger OLS, ADF

$p < 0.05$, $\beta > 0$); (3) Hurst exponent $H < 0.55$ confirming mean-reversion; (4) Ornstein–Uhlenbeck half-life between 2 and 156 weeks. Of 20 pairs tested, 15 are rejected at cointegration, 1 at the Hurst criterion, and **4 pass all filters**:

Pair	β	Half-life	H
LG30TRUU / LP01TREU	3.343	82 wk	0.508
LP01TREU / MXEU	0.101	64 wk	0.549
EMUSTRUU / LMBITR	0.775	25 wk	0.505
LMBITR / LUACTRUU	0.348	32 wk	0.535

Methodology

Threshold baseline. Z-score normalisation is applied using training mean and standard deviation only (no leakage). A long position is entered when $z < -2\sigma$ and a short when $z > +2\sigma$; both exit when z crosses zero. A 1-week execution delay and a 5-bar stabilising period are enforced. Transaction costs include 0.28% per leg and a short rental of 1%/52 per week.

LSTM anomaly detector. A SpreadLSTM model (PyTorch) is trained *exclusively* on $Y = 0$ weeks, learning the conditional distribution of normal spread dynamics. The architecture is: LSTM(h) $\rightarrow$ Dropout(0.2) $\rightarrow$ Dense(1). The anomaly score at time t is:

$$s(t) = |\hat{x}(t) - x(t)|, \quad \hat{x}(t) = f_{\theta}(x_{t-W:t-1})$$

where x is the standardised spread and W is the lookback window. A signal is generated when $s(t) > \tau$; the direction is determined by the sign of the error (actual < predicted $\Rightarrow$ long; actual > predicted $\Rightarrow$ short). The signal is held while $s(t) > \tau$.

Hyperparameter tuning. Nested time-series cross-validation with 3 expanding outer folds is used; the inner validation set is the last 20% of each outer-training window. A grid of 16 combinations per pair is searched: $W \in \{4, 8, 13, 26\}$, $h \in \{16, 32\}$, $\tau \in \{0.5, 1.0\}$. The optimisation criterion is the annualised Sharpe ratio (rf= 0, in-position weeks, $\times \sqrt{52}$). All scaling is fit on the training fold only.

Experimental Setup & Results

Split. Both models are trained on Jan 2000 – Dec 2019 (~1,040 weeks) and evaluated on Jan 2020 – Apr 2021 (~70 weeks), a period that coincides entirely with the COVID-19 market shock and subsequent recovery.

Results. Portfolio-level metrics are reported below. The portfolio Sharpe is the equally weighted average across the 4 pairs; ROI is the average per-pair return over the test window.

Metric	Threshold	LSTM
Portfolio Sharpe	0.930	2.596
Avg ROI	+1.37%	+10.23%
Avg Annual ROI	+1.03%	+7.60%
Total trades	2	14
Win rate	100%	64%
Δ Sharpe (LSTM – Thr.)	+1.666	

The threshold model generates only 2 trades: the $\pm 2\sigma$ threshold, calibrated on training variance, is rarely breached since the COVID shock drives spreads far beyond historical norms. The LSTM, by contrast, detects the anomalous March 2020 dislocation, enters mean-reversion positions, and profits from the subsequent recovery.

The star performer is **LP01TREU / MXEU** (European IG corporate bonds vs. MSCI Europe equities), which compressed severely in the crash and rebounded fast: test Sharpe of 4.01 and ROI of +40.38% over ~70 weeks.

Key Findings

1. OPTICS effectively reduces the search space.

Clustering on PCA-reduced returns reduces 861 candidate pairs to 20 economically coherent ones, with no manual domain filtering. Only 4 ultimately pass the full cointegration pipeline.

2. The LSTM is purpose-built for regime breaks.

Trained on normal-period spread dynamics, the anomaly score spikes sharply on the March 2020 crash. Mean-reversion positions entered during the dislocation capture the recovery: portfolio Sharpe = 2.596, Avg ROI = +10.23%.

3. The threshold model is robust but inert under COVID.

With only 2 trades generated, the $\pm 2\sigma$ strategy effectively disengages during the shock. Performance is positive (+1.37%) but nearly model-free.

4. CV reliability degrades without stress events in the validation fold.

Two of four pairs return a validation Sharpe of zero in inner CV (no dislocations in 2016–2019), rendering hyperparameter selection effectively random for those pairs. A robust design should ensure at least one CV fold covers a historical stress episode.

Conclusion

This work demonstrates that an LSTM novelty detector trained on normal-period spread dynamics can substantially outperform a classical threshold strategy during genuine regime breaks. OPTICS clustering, combined with a strict cointegration pipeline, provides an efficient and principled method for reducing the pair search space. The COVID-19 period serves as a natural stress test: the model catches the March 2020 dislocation cleanly and profits from the subsequent mean reversion. Future work should investigate rolling retraining, multi-pair joint modelling, and CV designs that guarantee stress-period coverage in at least one validation fold.